Lab 6 String, Numeric, and Set Operations

Creating Database and Tables

```
CREATE DATABASE IF NOT EXISTS school_db;
USE school_db;

DROP TABLE IF EXISTS Performance_Audit;
DROP TABLE IF EXISTS Primary_Students;
DROP TABLE IF EXISTS Secondary_Students;

CREATE TABLE Primary_Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100),
    marks INT,
    grade_level VARCHAR(10)
);

CREATE TABLE Secondary_Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100),
    marks INT,
    grade_level VARCHAR(10)
);

INSERT INTO Primary_Students (student_id, name, email, marks, grade_level) VALUES
(101, 'Sahasra Sharma', 'sahasra.s@school.edu', 88, 'Grade 5'),
(102, 'Shiya Verma', NULL, 92, 'Grade 5'),
(103, 'Raghav Kumar', 'raghav.k@school.edu', 74, 'Grade 6');

INSERT INTO Secondary_Students (student_id, name, email, marks, grade_level) VALUES
(104, 'Sahasra Sharma', 'sahasra.s@school.edu', 88, 'Grade 5'),
(105, 'Kavya Nair', 'kavya.n@school.edu', 95, 'Grade 7'),
(106, 'Rohan Das', 'rohan.d@school.edu', 65, 'Grade 8');
```

Pattern Matching (LIKE)

```
SELECT student_id, name, email
FROM Primary_Students
WHERE name LIKE 'A%' OR name LIKE 'S%';

SELECT student_id, name, email
```

```
FROM Primary_Students
WHERE email LIKE '%school%';
```

Output:

Output:

```
+------------+----------------+----------------------+
| student_id | name           | email                |
+------------+----------------+----------------------+
|        101 | Sahasra Sharma | sahasra.s@school.edu |
|        102 | Shiya Verma    | NULL                 |
+------------+----------------+----------------------+
+------------+----------------+----------------------+
| student_id | name           | email                |
+------------+----------------+----------------------+
|        101 | Sahasra Sharma | sahasra.s@school.edu |
|        103 | Raghav Kumar   | raghav.k@school.edu  |
+------------+----------------+----------------------+
```

Range Matching (BETWEEN / NOT BETWEEN)

```
SELECT student_id, name, marks
FROM Primary_Students
WHERE marks BETWEEN 75 AND 90;

SELECT student_id, name, marks
FROM Primary_Students
WHERE marks NOT BETWEEN 80 AND 95;
```

Output:

Output:

```
+------------+----------------+-------+
| student_id | name           | marks |
+------------+----------------+-------+
|        101 | Sahasra Sharma |    88 |
+------------+----------------+-------+
+------------+-------------+-------+
| student_id | name        | marks |
+------------+-------------+-------+
|        103 | Raghav Kumar |    74 |
+------------+-------------+-------+
```

List Matching (IN / NOT IN)

```
SELECT student_id, name, grade_level
FROM Primary_Students
WHERE name IN ('Sahasra Sharma', 'Raghav Kumar');
```

```
SELECT student_id, name, grade_level
FROM Primary_Students
WHERE grade_level NOT IN ('Grade 6', 'Grade 8');
```

Output:

Output:

```
+------------+----------------+-------------+
| student_id | name           | grade_level |
+------------+----------------+-------------+
|        101 | Sahasra Sharma | Grade 5     |
|        103 | Raghav Kumar   | Grade 6     |
+------------+----------------+-------------+
+------------+----------------+-------------+
| student_id | name           | grade_level |
+------------+----------------+-------------+
|        101 | Sahasra Sharma | Grade 5     |
|        102 | Shiya Verma    | Grade 5     |
+------------+----------------+-------------+
```

Existence Check (EXISTS)

```
SELECT p.student_id, p.name, p.grade_level
FROM Primary_Students p
WHERE EXISTS (
   SELECT 1
   FROM Secondary_Students s
   WHERE s.email = p.email
);
```

Output:

Output:

```
+------------+----------------+-------------+
| student_id | name           | grade_level |
+------------+----------------+-------------+
|        101 | Sahasra Sharma | Grade 5     |
+------------+----------------+-------------+
```

Set Operations (UNION / UNION ALL)

```
SELECT name, email, grade_level FROM Primary_Students
UNION
SELECT name, email, grade_level FROM Secondary_Students;

SELECT name, email, grade_level FROM Primary_Students
UNION ALL
SELECT name, email, grade_level FROM Secondary_Students;
```

Output:

Output:

```
| name           | email                | grade_level |
+----------------+----------------------+-------------+
| Sahasra Sharma | sahasra.s@school.edu | Grade 5     |
| Shiya Verma    | NULL                 | Grade 5     |
| Raghav Kumar   | raghav.k@school.edu  | Grade 6     |
| Kavya Nair     | kavya.n@school.edu   | Grade 7     |
| Rohan Das      | rohan.d@school.edu   | Grade 8     |
+----------------+----------------------+-------------+
+----------------+----------------------+-------------+
| name           | email                | grade_level |
+----------------+----------------------+-------------+
| Sahasra Sharma | sahasra.s@school.edu | Grade 5     |
| Shiya Verma    | NULL                 | Grade 5     |
| Raghav Kumar   | raghav.k@school.edu  | Grade 6     |
| Sahasra Sharma | sahasra.s@school.edu | Grade 5     |
| Kavya Nair     | kavya.n@school.edu   | Grade 7     |
| Rohan Das      | rohan.d@school.edu   | Grade 8     |
```